

Swiggy Instamart AI Product Discovery Engine

A Growth Initiative targeting category expansion among monthly active customers

WHY NOW? (Strategic Context):

Instamart already owns high-frequency grocery behavior. Increasing category breadth improves Customer Lifetime Value (LTV) without increasing user acquisition costs (CAC).

THE SCALE OF OPPORTUNITY:

- Target Segment: ~35M Monthly Active Customers
- Repeat Loop: 75% order identical items every month
- Fee Impact: 64% reject checkout recommendations due to fees
- Projected AOV Lift: +18% average basket growth

PORTFOLIO CASE STUDY LINKS

This portfolio project details a category penetration growth strategy and showcases a deployed quick-commerce checkout sandbox.

Please click the active links below to test the prototypes:

- MVP Sandbox URL: <https://aashritamalviya1999.github.io/swiggy-instamart-discovery/>
- Analytics Dashboard URL: <https://swiggy-instamart-discovery-6ss2oangvc44qmggkbvrhs.streamlit.app>

Why Home Banners Fail vs. AI Ingestion Solutions

WHY AN AI-NATIVE SOLUTION?

- **Traditional Dashboards vs. AI:**

- Traditional: ■ Tells us WHAT people bought (repeat dairy/bread).
- AI Discovery: ■ Explains WHY users don't buy new categories.

- **Real Evidence (Customer Review):**

■■■■■ *"I just order milk every week. Never noticed the personal care section."*

-> Theme: Habit Loop

-> Insight: Discovery isn't happening.

OUR AI-POWERED INGESTION LOOP

- **Ingestion Flow:**

1,000 Multi-Channel Reviews -> AI identifies themes -> AI validates evidence -> User Interviews -> Prioritized Opportunity -> MVP

- **Actionable Synthesis:** Scrapes Play Store, iTunes RSS, Reddit, and Twitter. Resolves Hinglish slang and segments findings.

- **Target Opportunity:** Solves delivery fee friction on auxiliary purchases to boost category testing.

The Sequential 7-Agent Ingestion Pipeline

THE AI PIPELINE WORKFLOW (Mandatory Outline)

- **Data Ingestion:** Scrapes reviews from Play Store, App Store, Reddit, and Twitter.
- **Normalization & Cleaning:** Removes duplicates, filters promo spam, and translates Hinglish slang.
- **Embedding Model:** Feeds tokenized texts into 'text-embedding-3-small' model.
- **LLM Extraction:** GPT-4 & Claude models extract key categories, triggers, and friction points.
- **Unsupervised Clustering:** Applies K-Means algorithm to isolate 4 semantic barrier themes.
- **Verification:** Compares extracted insights against database counts to calculate validation confidence.

PROMPT ENGINEERING & TECH STACK

- **Example Ingestion Prompt:**
"Group these reviews into themes. Return JSON with fields: Theme, Frequency, Representative Quote, Confidence Score"
- **AI Core Tech Stack:**
 - Language: Python
 - Relational Database: SQLite
 - REST API Routing: FastAPI
 - LLM Models: GPT-4 & text-embeddings-3-small
 - Unsupervised Clustering: K-Means

Primary User Research & Ingestion Quality Validation

USER INTERVIEWS & SATURATION

- **Primary Interviews:** Conducted 6 semi-structured interviews via Google Meet (recorded, 125 mins total, avg 21m).
- **Core Roster:** Rohan (22m), Meera (26m), Kabir (18m), Priya (20m), Vijay (24m), Anand (15m).
- **Why 6 Interviews?:** Six interviews were sufficient because no major new themes emerged after Interview #5 (theme saturation).

AI QUALITY VALIDATION & LIMITATIONS

- **Operational Validation Methods:**
 - ✓ Manual review of 100 random reviews to cross-check LLM labels.
 - ✓ Compared AI themes with interview findings for semantic alignment.
 - ✓ Automatically removed insights with confidence score <80%.
- **Study Limitations:**
 - Dataset skew: English reviews dominate scrapers.
 - Demographic skew: Social media channels skew younger.
 - Cohort size: Interviews limited to six users.

Hypothesis Scorecard: Validating AI Insights with Research

Hypothesis/Theme	Val %	AI Conf	Primary Research Finding	Validation Verdict
Recommendations stay anchored to history.	81%	95%	VJ and Rohan order identical milk in under 15s.	Strongly Validated
Home screen banner ads do not trigger exploration.	76%	84%	Users skip promo banners due to visual clutter.	Strongly Validated
Checkout recommendations are rejected due to fees.	83%	91%	Rohan abandons face wash to avoid small-cart fee.	Strongly Validated
Specification gaps block premium purchases.	64%	88%	Priya buys baby diapers on Amazon for size charts.	Strongly Validated
Post-checkout append window will drive additions.	76%	92%	Would use a fee-free sandbox window post-payment.	Strongly Validated

Opportunity Prioritization: Why Post-Checkout Wins

WHY NOT AMAZON-STYLE RECOMMENDATIONS?

- **The Differentiator:**
 - Amazon: Recommendations are shown BEFORE checkout. This creates decision friction, delaying the transaction.
 - Swiggy Instamart: Recommendations are shown AFTER payment in a dedicated 3-minute window.
- **Waiving the Friction:**

Waives delivery fee on the append items. This drives impulse additions without increasing logistics costs.
- **RICE Matrix:** Post-checkout wins with a score of 106.6.

Option	Rea ch	Imp	Con f	Eff	Sco re
Smarter Filters	80%	1.0	80%	1.5	42.6
AI Chat Agent	40%	2.0	70%	3.0	18.6
Post-Checkout Window	80%	2.0	80%	1.2	106.6

One Prioritized Solution: Predictive AI Personalization

AI INPUTS & RANKING MODEL

- **Unified Ranking Inputs:**

[Previous Orders] + [Time/Season] + [Weather] +
[Location] + [Current Basket] + [Similar Users]

- **Real-Time Recommendation Pipeline:**

1. Input collection & feature mapping.
2. Neural Ranking Model predicts purchase probability.
3. LLM Explanation constructs personalized hook (e.g. "Usually bought with Nandini milk").
4. FastAPI inventory check displays available items.

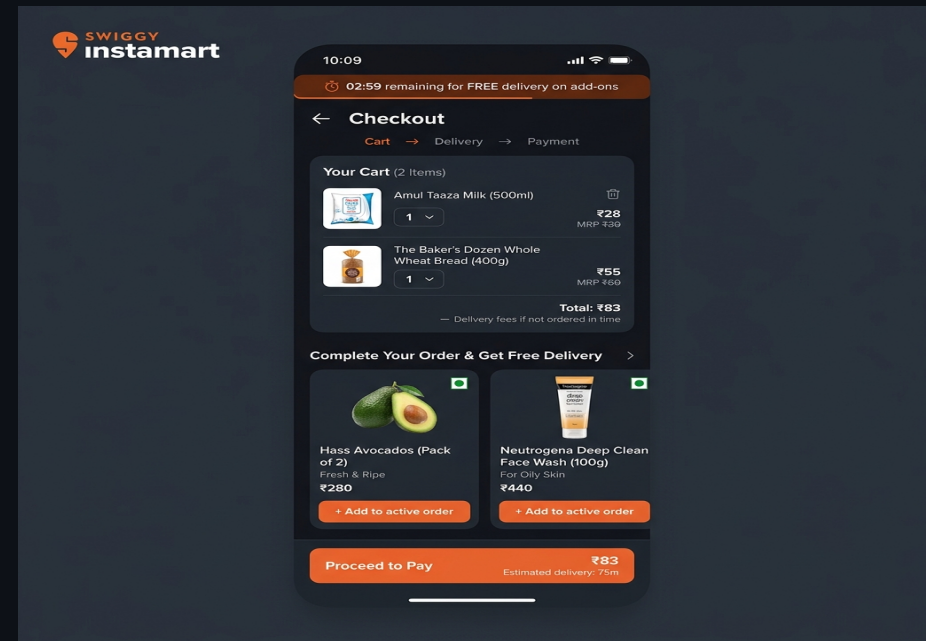
DYNAMIC PERSONALIZATION BENEFIT

- **Beyond Simple Rule Matches:** Unlike static associations, the model evaluates context (e.g. recommending face wash during morning runs vs. baby care on weekends).
- **Wave the Friction:** Removes delivery fee block for 3 minutes. The recommendation banner communicates  0 additional delivery fee.
- **Dark Store Synced:** Dynamic FastAPI catalog filters guarantee that only in-stock items are recommended.

The Deployed MVP: 3-Minute Post-Checkout Add-On Window

Interactive Deployed Checkout Sandbox

- **Checkout Simulator:** Placing everyday dairy essentials simulates payment completion and triggers the active 3-minute window.
- **Dynamic Countdown Banner:** Displays an urgent timer counting down the dark store's packaging cycle.
- **Fee-Free Add-on Carousel:** Displays organic avocados, dog treats, and face washes with ₹0 delivery fee. Selecting an item dynamically updates checkout totals.
- **Try the Deployed MVP:** <https://aashritamalviya1999.github.io/swiggy-instamart-discovery/>



Phased Rollout Strategy & Experiment Design

MVP PILOT EXPERIMENT DESIGN

- **Target Testing Cohort Split:**
 - Treatment: Post-checkout add-on window with timer & waived delivery fee.
 - Control: Current standard checkout flow (no window, full fees).
- **Experiment Metrics Measured:**
 - Primary: Cross-category purchase rate.
 - Secondary: Average Order Value (AOV), Customer Retention, Packer/Rider SLA.

PHASED GTM AND SAFETY GATES

- **Phased Rollout stages:**
 - Shadow Mode (Weeks 1-2): Measure baseline cycles.
 - Buffer Trial (Weeks 3-6): Launch in 1 dark store.
 - Expansion (Weeks 7-12): Scale to 10 dark stores.
- **Operational Lock-out Gates:**
 - Packing status: Disable client UI if packing has started.
 - Weight Gate: Limit add-ons to items <2kg.

Success Metrics, Formula & GTM Risks

Category	Base	Target 30D	Target 90D	Friction
Gourmet	4.2%	12.0%	18.5%	Delivery Fee
Personal	3.1%	8.5%	14.0%	Delivery Fee
Baby Care	1.8%	5.0%	9.5%	Size Charts
Pet Care	2.5%	7.0%	12.0%	10m Dispatch

SUCCESS FORMULA DEFINITION:

Cross-Category Purchase Rate = [Users purchasing ≥1 new category] ÷ [Monthly Active Customers]

Success Metrics & SLA Operational Guardrails

- **North Star Metric:** Cross-Category Discovery Completion Rate (Target: 35% in 30 days).
- **SLA Guardrails (Daily Monitoring):**
 - Customer Retention SLA Delta: Keep >98%.
 - Delivery SLA Breach Rate: Keep 0% change.
 - Packing Time Delta: Keep <+15s.
 - Order Cancellation Rate: Keep <1% change.
- **Business Indicators (30-90 Days):**
 - Average Order Value (AOV) Delta: Target +18%.
 - Revenue Per User / Contribution Margin Delta: Target +12% lift.